Caltrans Data Dictionary

Quality Assurance and Quality Control Process

Version [3.1] | [April 2026]

1. Purpose and Scope

This document describes the Quality Assurance & Quality Control process for the Caltrans Data Dictionary submission. It describes the two-stage review process, sets out what each stage checks, and defines what constitutes an acceptable submission. It also defines the deliverables required to successfully complete the process.

Document Suitability: This document is issued for review and comment. Once the process is fully agreed upon between all parties involved, the document will be reissued as an official process document.

2. Required Deliverables

#	Deliverable	Description
1	Data Dictionary Workbook	A completed version of the ISO-aligned workbook covering all object types within the consultant's scope. All 11 prescribed tabs must be present. No other workbook format or version will be accepted.
2	Information Requirements Workbook	The Information Requirements workbook. This is the baseline required to support both Stages 1 and 2 checks, along with the checklists shared below in Sections 5 and 6.
3	Sample Datasheets	A completed datasheet for each object type, produced using the prescribed user-friendly template format. Each datasheet must include a populated Value column demonstrating how the properties defined in the workbook would be populated for a representative asset instance. As agreed, the sample datasheets shall only be provided for Iteration 2.

3. Process Overview

The QA/QC process consists of two checks that run in parallel.

- Data Schema Check: Validates the Data Dictionary schema, compliance with ISO 23386, ISO 23387, ISO 12006 and ASCE 75-22 standards, and traceability of Information Requirements.

- Engineering Correctness Check: Validates the Information Requirements workbook. Validates whether objects have been defined correctly; properties defined for objects are sufficient, correct, and consistent with what is required for the purposes of design, construction, ~~operations and maintenance.~~

CONSULTANT
Submits Information Requirements Workbook
Prepares Data Dictionary Workbook
Prepares Data Sheet with Sample Values

V

PARALLEL REVIEW - Both checks run concurrently

DATA SCHEMA CHECK	ENGINEERING CORRECTNESS CHECK
Schema and structure validation ISO 12006 / 23386 / 23387 compliance IR traceability Attribute format checks Datasheet format verification	Validates Information Req. workbook Information Requirements property completeness Lifecycle coverage Property suitability review Datasheet content validation

Feedback from both checks issued together

4. Review Outcomes and Severity Ratings

Based on the data schema and engineering correctness checks, the submission will be marked as follows:

- **Accepted** - Submission accepted.
- **Requires Revision -** Submission is returned. All Fail items must be addressed before re-submission.

Feedback will be issued in the following formats.

- A completed Stage 1 checklist in Excel format, with the Result column (Pass / Fail / N/A) and Notes column populated for each check item, including screenshots or specific references to defects where applicable.
- A completed Stage 2 checklist in Excel format, with the Result column and Notes column populated for each check item, including specific engineering feedback and references where applicable.
- The overall review outcome: Accepted, Requires Revision.

Every checklist item carries a severity rating. Outcomes are determined as follows:

Severity	Meaning
Critical	Schema or structural issue compromising data model integrity or Standard compliance.
Advisory	A quality issue that should be resolved.

Outcome	Condition	Action
Accepted	Zero Critical failures. Zero Advisory flags.	Submission accepted.
Requires Revision	One or more Critical failures, **or** one or more Advisory flags.	Submission is returned. All Fail items must be addressed before re-submission.

5. Stage 1: Data Schema Checks

This stage verifies that the data dictionary workbook and sample datasheets are structurally sound, complete, and internally consistent. Stage 1 does not assess engineering correctness - that is the purpose of Stage 2.

Submission Reference		**Consultant Name**	
Submission Date		**Review Date**	
Reviewer Name		**Reviewer Role**	
Checklist Version	1.0		

Ref	Category	Check	Severity	Result	Notes
Structural Integrity					
S1-01	Structural Integrity	All 11 prescribed workbook sheets are present and correctly named (including trailing spaces where applicable): READ_ME, CONTROLLED_LISTS, ID_REGISTER, OBJECTS_12006, OBJECT_RELATIONSHIPS_12006, TEMPLATES_23387, PROPERTIES_23386, PROPERTY_GROUPS_23386, GROUP_PROPERTY_MEMBERSHIP, TEMPLATE_GROUP_REFS, TEMPLATE_PROPERTY_REFS.	**Critical**	Pass / Fail / N/A	
S1-02	Structural Integrity	PROPERTIES_23386 contains all 41 prescribed columns: the 40 ISO 23386 Table 1 attributes (PA001–PA040) plus the ISO 12006-derived Property Key column.	**Critical**	Pass / Fail / N/A	
S1-03	Structural Integrity	PROPERTY_GROUPS_23386 contains all 24 prescribed columns: the 23 ISO 23386 Table 2 attributes (GA001–GA023) plus the ISO 12006-derived Group Key column.	**Critical**	Pass / Fail / N/A	
S1-04	Structural Integrity	All column headers in PROPERTIES_23386 and PROPERTY_GROUPS_23386 match the prescribed ISO 23386 Table 1 and Table 2 attribute names exactly. No abbreviations, reformatting, or variations are accepted.	**Critical**	Pass / Fail / N/A	
S1-05	Structural Integrity	Every Object_Key follows the prescribed URN format: urn:<Org>:obj:<Name>. Applies to OBJECTS_12006 and all cross-reference fields in OBJECT_RELATIONSHIPS_12006.	**Critical**	Pass / Fail / N/A	
S1-06	Structural Integrity	Every Property Key follows the prescribed URN format: urn:<Org>:property:<Name>. Applies to PROPERTIES_23386, GROUP_PROPERTY_MEMBERSHIP, and TEMPLATE_PROPERTY_REFS.	**Critical**	Pass / Fail / N/A	
S1-07	Structural Integrity	Every Group_Key follows the prescribed URN format: urn:<Org>:propertygroup:<Name>. Applies to PROPERTY_GROUPS_23386, GROUP_PROPERTY_MEMBERSHIP, and TEMPLATE_GROUP_REFS.	**Critical**	Pass / Fail / N/A	
S1-08	Structural Integrity	Every Template_Key follows the prescribed URN format: urn:<Org>:dt:<Name>. Applies to TEMPLATES_23387, TEMPLATE_GROUP_REFS, and TEMPLATE_PROPERTY_REFS.	**Critical**	Pass / Fail / N/A	
S1-09	Structural Integrity	All Globally_unique_identifier values are properly formatted UUIDs and are unique within each sheet.	**Critical**	Pass / Fail / N/A	
S1-10	Structural Integrity	All objects, properties, groups, and templates are registered in ID_REGISTER.	**Critical**	Pass / Fail / N/A	

Completeness					
S1-11	Completeness	All unconditionally mandatory ISO 23386 Table 1 attributes are populated for every property entry in PROPERTIES_23386: PA001, PA002, PA003, PA006, PA007, PA009, PA010, PA015, PA016, PA017, PA021, PA024, PA027, PA030, and PA033. Descriptions (PA018) and Units (PA033) are mandatory for every property without exception. Non-physical properties must use 'without' for PA027 and 'unitless' for PA033. Dimension (PA028) is mandatory when Physical_Quantity is not 'without'. PA039 (List of possible values) is mandatory where Data_type is 'Enumeration'. PA029 (Method of measurement) is mandatory where a measurement method is applicable. PA014 (Relation_of_the_property_identifiers_in_the_interconnected_data_dictionaries) should be mandatory for any property where an equivalent exists in an external interconnected data dictionary (such as IFC) to ensure semantic interoperability.	Critical	Pass / Fail / N/A	
S1-12	Completeness	All unconditionally mandatory ISO 23386 Table 2 attributes are populated for every property group entry in PROPERTY_GROUPS_23386: GA001, GA002, GA003, GA005, GA006, GA007, GA009, GA010, GA015, GA016, GA017, GA019 and GA022.	Critical	Pass / Fail / N/A	
S1-13	Completeness	Every template in TEMPLATES_23387 has at least one property assigned in TEMPLATE_PROPERTY_REFS.	Critical	Pass / Fail / N/A	
S1-14	Completeness	Every template in TEMPLATES_23387 has at least one property group assigned in TEMPLATE_GROUP_REFS.	Critical	Pass / Fail / N/A	
S1-15	Completeness	Templates with fewer than 10 properties assigned in TEMPLATE_PROPERTY_REFS are flagged. Consultant must provide written justification.	Advisory	Pass / Fail / N/A	
S1-16	Completeness	Templates missing property groups that would reasonably be expected for the asset type are flagged. Consultant must confirm omission is intentional.	Advisory	Pass / Fail / N/A	
S1-17	Completeness	Every property group in PROPERTY_GROUPS_23386 has at least one member property in GROUP_PROPERTY_MEMBERSHIP.	Critical	Pass / Fail / N/A	
S1-18	Completeness	Every property in PROPERTIES_23386 belongs to at least one property group in GROUP_PROPERTY_MEMBERSHIP.	Critical	Pass / Fail / N/A	
S1-19	Completeness	IFC entity, IFC predefined type, and OmniClass classification attributes are populated for each object type in OBJECTS_12006.	Advisory	Pass / Fail / N/A	
S1-20	Completeness	AASHTOWare pay item codes are populated for each object in OBJECTS_12006. Correctness of assigned codes is validated at Stage 2.	Advisory	Pass / Fail / N/A	
S1-21	Completeness	Where Data_type is 'Enumeration', each entry in PA039 (List_of_possible_values_in_language_N) must follow the ISO 23386 prescribed pair format: (value | language_code). A plain list of values without language codes does not conform to the standard. Example of correct format: (New Installation | en-US) || (Repair | en-US) || (Removal | en-US).	Advisory	Pass / Fail / N/A	
IR Traceability					
S1-22	IR Traceability	Every property listed in the Information Requirements workbook for each object type is present in PROPERTIES_23386. No IR-required property is absent from the data dictionary workbook.	Critical	Pass / Fail / N/A	

S1-23	IR Traceability	Every property defined in PROPERTIES_23386 for each object type appears in the corresponding sample datasheet. No workbook property is absent from its datasheet.	**Critical**	Pass / Fail / N/A	
S1-24	IR Traceability	Where a property has a known equivalent in an external reference dictionary (such as IFC or the buildingSMART Data Dictionary), PA014 (Relation_of_the_property_identifiers_in_interconnected_data_dictionaries) must be populated with the corresponding external identifier. Properties with no known external equivalent may leave PA014 blank. Where cross-dictionary linkage is expected for the asset class and PA014 is consistently unpopulated across all properties, this is flagged.	**Advisory**	Pass / Fail / N/A	
Consistency					
S1-25	Consistency	Only the four permitted relationship types are used in OBJECT_RELATIONSHIPS_12006: is_type_of, is_part_of, has_part, has_type. Any other value constitutes a failure.	**Critical**	Pass / Fail / N/A	
S1-26	Consistency	Composition relationships (has_part / is_part_of) are not conflated with classification relationships (has_type / is_type_of), consistent with ISO 12006-2 Sections 4.2–4.4.	**Critical**	Pass / Fail / N/A	
S1-27	Consistency	All cross-sheet GUID references resolve to valid entries in their respective source sheets.	**Critical**	Pass / Fail / N/A	
S1-28	Consistency	Data_type values are drawn from the agreed controlled list in CONTROLLED_LISTS and are consistent across all packages: String, Real, Date, Boolean, Integer, Enumeration, GUID.	**Critical**	Pass / Fail / N/A	
~~**S1-29**~~	~~Consistency~~	~~Units values are checked against ISO 23386 Annex C Tables C.1 and C.2. Non-SI units are flagged for consultant written justification and carried to Stage 2.~~	~~**Advisory**~~	~~Pass / Fail / N/A~~	
S1-30	Consistency	Where Physical_Quantity is not 'without', the Dimension field follows ISO 80000 format: seven space-separated exponents in the order L M T I Θ N J.	**Advisory**	Pass / Fail / N/A	
S1-31	Consistency	No entry in OBJECTS_12006 represents a non-physical concept. All objects are tangible physical assets or legitimate logical asset constructs.	**Critical**	Pass / Fail / N/A	
S1-32	Consistency	All data templates in TEMPLATES_23387 are assigned to physical or logical asset objects. Templates must not be assigned to classification categories.	**Critical**	Pass / Fail / N/A	
S1-33	Consistency	For every property group assigned to a template in TEMPLATE_GROUP_REFS, any absent member properties in TEMPLATE_PROPERTY_REFS are flagged. Consultant must confirm omissions are intentional.	**Advisory**	Pass / Fail / N/A	
S1-34	Consistency	No duplicate property entries exist in PROPERTIES_23386 (identical or near-identical name and definition with different GUIDs/URNs).	**Critical**	Pass / Fail / N/A	
S1-35	Consistency	Properties common to multiple asset types are reused by GUID reference and not redefined. Semantic duplicates across packages are flagged as Advisory.	**Advisory**	Pass / Fail / N/A	
S1-36	Consistency	Every entry in PROPERTY_GROUPS_23386 must have GA022 (Category_of_group_of_properties) populated with one of the five ISO 23386 permitted values: Domain, Class, Composed property, Alternative use, or Reference document. Any other value, or a blank entry, constitutes a Critical failure. Use Domain for groups representing areas of practice or discipline; Class for groups specific to a defined object class; Reference document for groups tied to an external publication or standard;	**Critical**	Pass / Fail / N/A	

		Composed property for groups where all member properties must be populated together to carry meaning; and Alternative use as a last resort.			
S1-37	Consistency	Where PA028 (Dimension) is populated, the dimension vector must be consistent with the physical quantity stated in PA027 (Physical_Quantity). Reviewers must cross-check the dimension vector against ISO 80000 expected values for the stated quantity. Common expected vectors: Length = 1 0 0 0 0 0 0; Area = 2 0 0 0 0 0 0; Volume = 3 0 0 0 0 0 0; Mass = 0 1 0 0 0 0 0; Force = 1 1 -2 0 0 0 0; Pressure = -1 1 -2 0 0 0 0. A vector that does not correspond to the stated physical quantity is flagged as Advisory.	**Advisory**	Pass / Fail / N/A	
S1-38	Consistency	Where a property in PROPERTIES_23386 shares an identical name (PA016) and definition (PA017) with an existing property in the shared property library, the GUID (PA001) must also match. Reproducing an existing property under a new GUID rather than referencing the original GUID constitutes a Critical failure. Semantic reuse must be implemented as structural reuse.	**Critical**	Pass / Fail / N/A	
S1-39	Consistency	Source_Cardinality and Target_Cardinality must be populated for every relationship entry in OBJECT_RELATIONSHIPS_12006. Blank cardinality fields are a Critical failure. Cardinality must be expressed in standard notation (e.g. 0..1, 1..1, 0..*, 1..*).	**Critical**	Pass / Fail / N/A	
S1-40	Consistency	The object hierarchy defined in OBJECTS_12006 and OBJECT_RELATIONSHIPS_12006 must be coherent. No missing intermediate classification levels are present that would leave objects without a classification parent where one is logically required. All physical component parts of a parent object are accounted for in the submitted scope or explicitly excluded with written justification. Identified hierarchy gaps are flagged as Advisory.	**Advisory**	Pass / Fail / N/A	
S1-41	Consistency	The Described_Object_Key in every TEMPLATES_23387 entry must resolve to a valid entry in OBJECTS_12006. A template whose Described_Object_Key does not correspond to any registered object constitutes a Critical failure.	**Critical**	Pass / Fail / N/A	
Datasheet - Format					
S1-42	Datasheet - Format	A sample datasheet has been submitted for each object type in the prescribed template format.	**Critical**	Pass / Fail / N/A	
S1-43	Datasheet - Format	A Value column is present immediately after Names_in_language_N and labelled exactly as 'Value'.	**Critical**	Pass / Fail / N/A	
S1-44	Datasheet - Format	Property group section headers in the datasheet mirror the grouping structure in the prescribed template.	**Critical**	Pass / Fail / N/A	
S1-45	Datasheet - Format	All properties referenced in the datasheet are defined in PROPERTIES_23386. No properties are introduced that are absent from the workbook.	**Critical**	Pass / Fail / N/A	
S1-46	Datasheet - Format	No Value cells are blank. Where a value is unavailable, one of the following is used: N/A, Not collected, Unknown.	**Advisory**	Pass / Fail / N/A	
S1-47	Datasheet - Format	Units in the Value column are consistent with those defined for each property in PROPERTIES_23386.	**Advisory**	Pass / Fail / N/A	

Stage 1 Outcome

Total Critical Failures	
Total Advisory Flags	
Overall Outcome	☐ Accepted ☐ Accepted with Comments ☐ Requires Revision
Reviewer Signature	
Date Signed	

6. Stage 2: Engineering Correctness Review

Caltrans engineers conduct Stage 2. It assesses whether submitted objects and properties are technically correct, sufficiently complete, and appropriate for Caltrans purposes. Approval authority rests solely with Caltrans.

Caltrans engineers are not expected to navigate or interpret the data dictionary workbook directly. The primary reference documents for Stage 2 are:

- The Information Requirements workbook is the authoritative baseline for what properties should exist and how they should be represented.
- The sample datasheets - the primary view of property names, definitions, units, data types, and sample values.

One instance of this checklist must be completed for each object type. Where a submission covers multiple object types, each receives its own checklist, header block, and sign-off.

Submission Reference		**Consultant Name**	
Submission Date		**Review Date**	
Reviewer Name		**Reviewer Discipline**	
Object Name		**Object GUID**	
Object URN		**Asset Class**	
Checklist Version			

Ref	Category	Check	Severity	Result	Notes
Objects & Relationships					
S2-01	Objects & Relationships	The object type is within the agreed scope of this submission and is listed in the approved scope schedule.	Critical	Pass / Fail / N/A	
S2-02	Objects & Relationships	The object name and definition are clear, appropriate for Caltrans, and sufficient to distinguish this object from related asset types.	Advisory	Pass / Fail / N/A	

S2-03	Objects & Relationships	The object is defined at the correct level of detail for Caltrans - neither too granular nor too coarse - reflecting how this asset class is managed in practice.	**Critical**	Pass / Fail / N/A	
S2-04	Objects & Relationships	The breakdown of components and subtypes shown in the Object Map reflects how Caltrans structures and manages this asset class. Components and subtypes are correctly distinguished (TBC if Object Map will be a deliverable).	**Critical**	Pass / Fail / N/A	
S2-05	Objects & Relationships	The IFC entity type and OmniClass classification assigned to this object appear correct and appropriate for this asset type.	**Critical**	Pass / Fail / N/A	
S2-06	Objects & Relationships	The AASHTOWare pay item codes assigned to this object correspond to the correct standard Caltrans pay items for this asset type.	**Critical**	Pass / Fail / N/A	
IR Alignment					
S2-07	IR Alignment	All properties listed in the Information Requirements workbook for this object type are present and correctly represented in the submission. No IR-required property is missing or incorrectly scoped.	**Critical**	Pass / Fail / N/A	
S2-08	IR Alignment	Properties derived from the Information Requirements workbook are correctly represented - names, definitions, and intended use are consistent with the intent of the IR.	**Critical**	Pass / Fail / N/A	
Properties - Lifecycle Sufficiency					
S2-09	Lifecycle Sufficiency	Properties assigned to this object are sufficient to support the design phase of the asset lifecycle. All data required to design the asset - including geometric, structural, material, and performance properties - is represented.	**Critical**	Pass / Fail / N/A	
S2-10	Lifecycle Sufficiency	Properties assigned to this object are sufficient to support construction and installation. All data required to construct, install, and inspect the asset is represented.	**Critical**	Pass / Fail / N/A	
~~**S2-11**~~	~~Lifecycle Sufficiency~~	~~Properties assigned to this object are sufficient to support operations and maintenance. All data required to operate, inspect, maintain, and manage the asset is represented.~~	~~**Critical**~~	~~Pass / Fail / N/A~~	
S2-12	Lifecycle Sufficiency	No significant gaps in the property set are identified beyond the lifecycle checks above.	**Critical**	Pass / Fail / N/A	
Properties - Technical Correctness					
S2-13	Technical Correctness	For each property, the physical quantity being measured is the correct and most operationally useful quantity for Caltrans purposes.	**Advisory**	Pass / Fail / N/A	

S2-14	Technical Correctness	The units defined for each property are operationally appropriate for how Caltrans collects, records, and uses this data in practice.	**Advisory**	Pass / Fail / N/A	
S2-15	Technical Correctness	The data type assigned to each property is appropriate for how the value is recorded and used in Caltrans practice (e.g., a condition rating using defined categories should be Enumeration, not Real).	**Advisory**	Pass / Fail / N/A	
S2-16	Technical Correctness	Where a method of measurement is defined for a property, it is consistent with Caltrans inspection and data collection practice. Where no method is defined for a property that operationally requires one, this is flagged.	**Advisory**	Pass / Fail / N/A	
Properties - General					
S2-17	Properties - General	Property names and definitions, as shown in the sample datasheet, are clear, unambiguous, and sufficient for consistent interpretation by Caltrans engineers and field data collectors. Definitions describe what is being measured, not merely restate the property name.	**Advisory**	Pass / Fail / N/A	
S2-18	Properties - General	Where a property has a list of possible values, those values are technically appropriate, sufficiently complete, and consistent with Caltrans standards and operational practice.	**Advisory**	Pass / Fail / N/A	
S2-19	Properties - General	Property groups are appropriate for this asset type and consistent in naming and structure with those used across other asset type packages. NOTE: Only applicable to reviewers overseeing the entire asset list. Reviewers assessing a single object type should record N/A.	**Advisory**	Pass / Fail / N/A	
Datasheet - Content Review					
S2-20	Datasheet - Content Review	Sample values in the Value column are realistic and representative of actual asset data for this object type.	**Advisory**	Pass / Fail / N/A	
S2-21	Datasheet - Content Review	The populated datasheet demonstrates that the data model would work in practice - properties are at the right level of detail, values are meaningful, and the overall record represents a credible, usable asset instance.	**Critical**	Pass / Fail / N/A	
S2-22	Datasheet - Content Review	No properties in the datasheet are assessed as unnecessary, irrelevant, or inappropriate for this asset type.	**Advisory**	Pass / Fail / N/A	

Object Outcome and Sign-off	
Total Critical Failures	
Total Advisory Flags	

Object Outcome	☐ Accepted ☐ Accepted with Comments ☐ Rejected
Caltrans Authorizer Name	
Caltrans Authorizer Signature	
Authorization Date	

7. Do's and Don'ts

The following summarises the most important practices for producing compliant submissions.

DO	DO NOT
• Use only the four permitted relationship types: is_type_of, is_part_of, has_part, has_type	• Invent relationship types outside the four permitted values
• Assign properties to objects via templates, not directly to the object	• Define non-physical concepts (informational or administrative constructs) as objects in OBJECTS_12006
• Ensure all 40 Table 1 and 23 Table 2 attribute columns are present and correctly named in PROPERTIES_23386 and PROPERTY_GROUPS_23386	• Submit a workbook with missing attribute columns or incorrectly named column headers
• Use SI units as default. Cross-check against ISO 23386 Tables C.1 and C.2. Justify any non-SI units in writing	• Use non-SI units without written justification citing the applicable Caltrans or project standard
• Reuse existing properties by GUID reference across asset types rather than redefining them	• Redefine properties that already exist in the shared property set - reference them by GUI instead
• Populate the Value column in the datasheet with realistic sample values. Use N/A, Not collected, or Unknown if a value is unavailable	• Leave Value column cells blank in the datasheet
• Submit both the completed workbook and sample datasheets using the current prescribed versions	• Submit based on a superseded workbook or template version
• Populate all 15 unconditionally mandatory Table 1 and 13 Table 2 attributes for every property and property group entry	• Conflate classification relationships (has_type / is_type_of) with composition relationships (has_part / is_part_of)

Annex A: Sample Datasheet Format

SAMPLE DATASHEET - OBJECT INFORMATION			
Object Name	[Object Name]	**Object GUID**	[xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx]
Object URN	[urn:<Org>:obj:<ObjectName>]		
IFC Entity	[IFC Entity]	**IFC Predefined Type**	[IFC Predefined Type]
IFC Schema Version	[IFC Schema Version]	**OmniClass Code**	[OmniClass Code]
OmniClass Table	[OmniClass Table]	**OmniClass Edition**	[OmniClass Edition]
AASHTOWare Pay Item Code(s)	[Pay Item Code(s)]		
Submission Reference		**Submission Date**	
Consultant		**Prepared By**	

Names_in_language_N (PA016)	Definitions_in_language_N (PA017)	Value	Units (PA033)	Physical_Quantity (PA027)	Data_type (PA030)	List_of_possible_values_in_language_N (PA039)	Dimension (PA028)	Method_of_measurement (PA034)
Property Group 1								
Property 1								
Property 2								
Property 3								
Property 4								
Property 5								
Property Group 2								
Property 6								
Property 7								
Property 8								

Annex B: Objects Map (for discussion whether this is needed)

This appendix defines the requirements for the Object Map and Relationship Map (Deliverable 4). This gives Caltrans engineers a clear visual reference of how submitted objects are structured and related, without requiring them to navigate the data dictionary workbook. They are the primary reference for Stage 2 check S2-04.

C.1 Format 1: Objects Map

The objects map shows all objects submitted within the package and all relationships between them in a single diagram.

Minimum content requirements:

- All objects submitted in OBJECTS_12006 for the package, identified by their object name.
- All composition relationships (has_part / is_part_of) shown as solid lines with a directional arrow from parent to component.
- All classification relationships (has_type / is_type_of) shown as dashed lines with a directional arrow from category to subtype.
- A legend clearly identifying the line style used for composition vs classification relationships.
- Object names shown in full. Object URNs shown in smaller text below each object name.
- A title block showing submission reference, consultant name, asset class, and submission date.
- Objects should be arranged hierarchically, with parent objects above their components and categories above their subtypes.